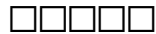


PHENIKAA UNIVERSITY

PHENIKAA SCHOOL OF COMPUTING



COURSE: SOFTWARE ARCHITECTURE

TOPIC:

QR CODE-BASED IN-TABLE FOOD ORDERING SYSTEM REPORT

LAB 1 – REQUIREMENTS ELICIATION & MODELING DOCUMENT:

SOFTWARE ARCHITECTURE

Code Course : CSE703110-1-3-25

Class : N01

Instructor : THS. Vũ Quang Dũng

Group : 1

Members : 1. Nguyễn Đình Trung - 23010701

2. Vũ Anh Nam - 23010886

3. Đặng Minh Hoàng – 23010445

4. Nguyễn Phan Quang Minh - 23010530

Table of content

1. Abstract.....	3
2. Requirements Elicitation.....	3
3. Use Case Modeling	5
4. Conclusion & Reflection.....	13

1. Abstract

a. Problem Description

The traditional operational workflow in current restaurants relies heavily on manual processes, ranging from paper menu distribution and order-taking to kitchen communication, bill printing, and counter checkout. This model introduces major operational bottlenecks, particularly during peak hours, including prolonged customer wait times, a high rate of human-induced order errors, expensive labor overhead, and static menus that lack real-time updates. To thoroughly address these limitations, the '**QR Code Food Ordering System**' project is developed to shift the control of browsing, ordering, and online transactions directly to the customer's smartphone. This digital solution eliminates unnecessary intermediary steps and automates cross-department coordination, thereby optimizing operational costs for the restaurant and significantly enhancing the overall dining experience for consumers.

b. Description of Lab 01

This lab focuses on the first phase of the software architecture process: requirements elicitation and system modeling. The main objective is to identify key functional and non-functional requirements of the selected system, determine the primary actors, and extract architecturally significant requirements (ASRs) that will influence architectural decision-making in later stages. A UML Use Case Diagram is constructed to model the system boundary, actors, and high-level interactions, providing a visual foundation for understanding the external behavior of the system. The outcomes of this lab establish the groundwork for architectural design activities in subsequent labs.

This foundation establishes the scope and the critical architectural drivers that will be addressed in subsequent labs, particularly the design of the Layered Architecture in next Lab 2.

2. Requirements Elicitation

a. Actors

Actor	Role / Responsibilities	Interaction With System
Customer	Acts as the primary end-user visiting the restaurant. Responsible for browsing the menu, selecting food items, managing choices, and completing payments.	Scans table QR code to view the menu. Searches, filters, and adds items to cart. Places orders directly to the kitchen. Performs online payment or requests cash payment.
Restaurant Staff	Responsible for receiving order details and preparing dishes accurately. Ensures food is delivered to the correct table and assists customers.	Logs into the staff app to receive real-time orders. Updates food preparation status (Preparing/Ready). Manages table service and availability.
Administrator	Responsible for managing the digital menu, restaurant configuration, and staff accounts. Monitors daily business performance and financial revenue.	Accesses the central administrative dashboard. Manages digital menu (items, prices, stock). Generates table QR codes and layout. Views and exports sales/revenue reports
Payment Gateway	Responsible for verifying accounts and ensuring safe, encrypted monetary transactions.	Receives payment requests and invoice details. Processes the transaction from customer accounts. Returns success/failed status to the system.

b. Functional Requirements (FRs)

ID	Functional Requirement	Priority
FR-01	Automatically recognizes the table and displays the digital menu instantly without registration.	High

FR-02	Allows customers to search items by name and filter by categories (mains, drinks,...).	Medium
FR-03	Allows customers to add, modify quantity, delete items, and add special kitchen notes.	High
FR-04	Sends the cart items directly to the kitchen display and generates a unique Order ID.	High
FR-05	Supports online payments (MoMo, VNPay) or cash payment requests to alert servers.	High

c. Non-Functional Requirements (NFRs)

ID	Attribute	Description	Impact
NFR-01	Performance (Latency)	The digital menu and food images must fully load within less than 2 seconds after the QR scan. The shopping cart update latency must be under 500ms.	High
NFR-02	Security (Integrity)	All order data, payment information, and transaction statuses must be protected against unauthorized alteration or data leakage during transmission.	Critical
NFR-03	Reliability (Availability)	The platform must guarantee a 99.9% operational uptime during active restaurant business hours (7:00 AM - 11:00 PM).	Critical
NFR-04	Usability	The web interface must be intuitive, minimal, and fully responsive across various mobile screen sizes without requiring app installation.	Medium

d. Architecturally Significant Requirements (ASRs)

ID	Quality Attribute	Description	Impact
ASR-01	Modifiability	Easily switch or add new payment gateways (e.g., MoMo, VNPay) without modifying the core ordering logic.	Isolate the payment feature into a separate, decoupled module to prevent code regression in other functions.
ASR-02	Security	Prevent client-side price manipulation and secure transaction data using standard HTTPS.	Enforce Server-side validation for all financial and cart calculations; never trust raw data sent from the browser.

ASR-03	Scalability	Handle concurrent order requests from multiple tables during peak hours without system freezing.	Keep application services stateless and optimize database queries to prevent processing bottlenecks.
---------------	--------------------	--	--

3. Use Case Modeling

a. Objective of the Use Case Model

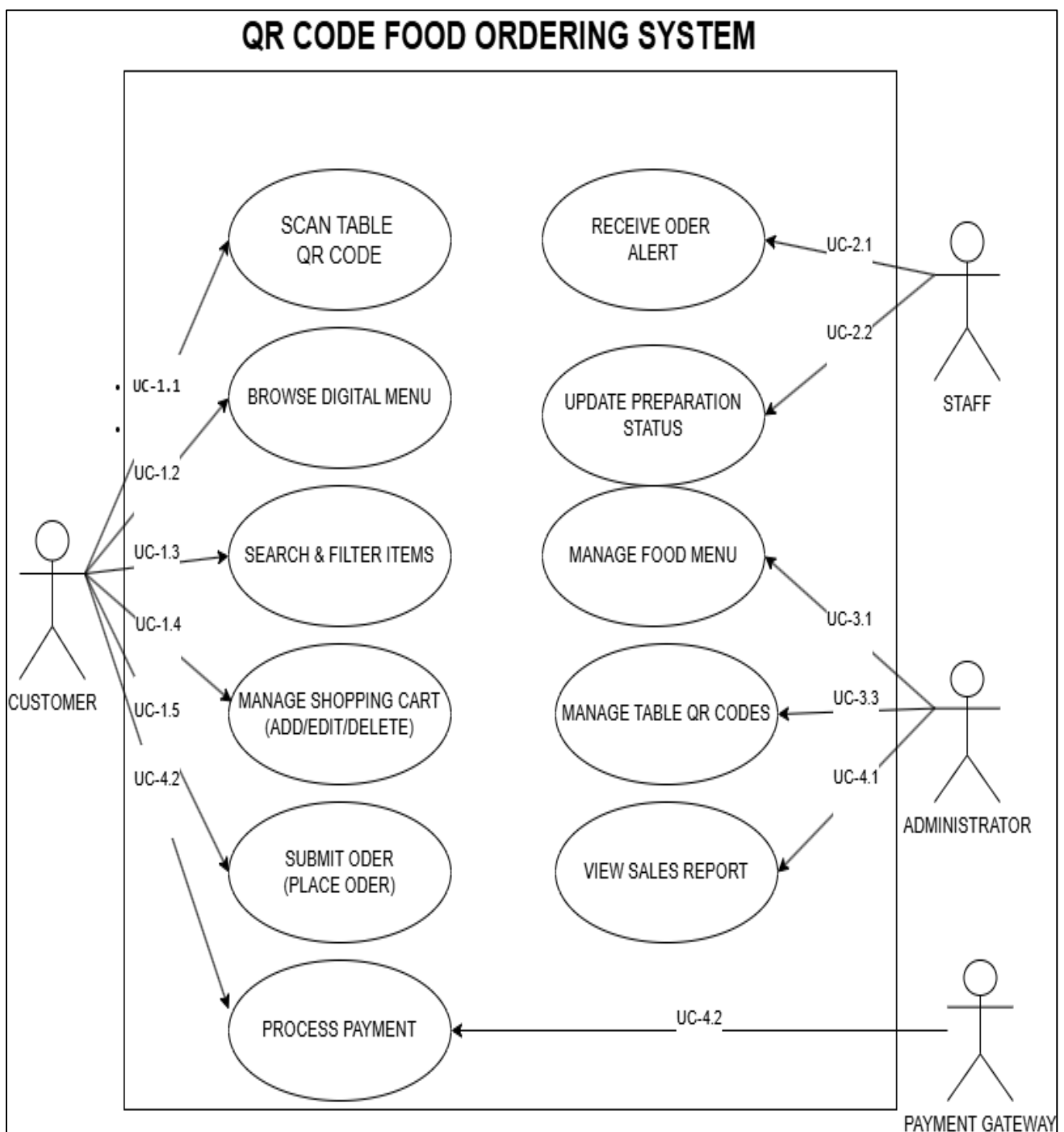
The primary objective of the Use Case Model is to clearly define the system boundary and specify the functional scope of the QR Code Food Ordering System from the users' perspective. By visually mapping the interactions between external actors (such as Customers, Staff, Admins, and Payment Gateways) and the system's core functions, the model establishes a clear functional blueprint for the project. For a demo-scale application, this model serves as an essential communication tool to align requirements among team members, while providing a foundational basis for designing the system architecture, detailing sequence flows, and structuring subsequent testing phases.

b. System Actors

The system interacts with four primary actors:

- Customer: Scans QR, browses the menu, manages the cart, places orders, and pays online.
- Restaurant Staff: Receives real-time table orders and updates food preparation status.
- Administrator: Manages the digital menu, prices, table QR codes, and views sales reports.
- Payment Gateway: Processes financial transactions and returns payment status.

c. Overview Use Case Diagram



d. Use Case List by Functional Group

UC1: Customer In-Table Ordering Management

UC-1.1 Browse e-Menu & Manage Cart	
Description	Customer scans the table's QR code to view categories, active items, customize options, and manage their local shopping cart.
Trigger	Customer opens the browser via table QR code or digital menu URL.
Preconditions	The system, item catalog, and database are available.
Postconditions	Selected items and modifications are successfully saved to the active local cart instance.
Normal Flows	1) Customer accesses the digital menu interface. 2) System displays active menu categories (e.g., Coffee, Smoothie) and available items. 3) Customer searches or filters items. 4) Customer adjusts item quantities, inputs special modifications, and adds them to the cart.
Alternative Flows	A1) Customer uses category tabs to jump directly to specific items. A2) Customer clicks the "Edit" link within the cart to return to the item details view for modification.
Exceptions	E1) Invalid or unmapped QR code scanned -> system displays error view. E2) System/DB connection failure.
Priority	High
Frequency of Use	High
Business Rules	Inactive or out-of-stock items must be hidden or greyed out; quantities cannot be negative.
Assumptions	Menu details include item code, name, price, category_id, availability status, and metadata.

UC-1.2 Place In-Table Order

Description	Customer reviews the finalized cart items, selects a payment method, and submits the order to the kitchen.
Trigger	Customer clicks the "Đặt đơn" (Place Order) button within the shopping cart view.
Preconditions	Shopping cart must contain at least one valid item; Table ID must be authenticated.
Postconditions	Order data is locked and stored in the database; customer cart is cleared; real-time order alert is pushed to the staff dashboard.
Normal Flows	1) Customer opens the checkout/shopping cart view. 2) System displays items, itemized costs, total price, and payment selections. 3) Customer selects a payment method (Cash or Bank Transfer) and submits the order. 4) System validates current stock, processes submission, and returns an "Order Placed Successfully" temporary toast.
Alternative Flows	A1) Customer modifies item counts or removes rows prior to confirmation. A2) Customer cancels checkout and goes back to the homepage.
Exceptions	E1) Item stock drops to zero during checkout -> prompt user to adjust cart. E2) Order creation transaction fails -> show connection error toast.
Priority	High

Frequency of Use	High
Business Rules	Each submission must explicitly bind to a verified Table ID; payment method field cannot be blank.
Assumptions	Order metadata includes order_id, table_id, timestamp, total_amount, payment_status, and order_status.

UC-1.3 Query Tableside Assistance / History

Description	Customer triggers an assistance alert or checks their current table's order queue state.
Trigger	Customer clicks "Gọi nhân viên" (Call Staff) or "Món đã gọi" (Order History) on the header navigation bar.
Preconditions	Customer session with a valid Table ID is active.
Postconditions	An interactive overlay modal or history list view is rendered based on user action.

Normal Flows	1) Customer taps the assistance or history action trigger. 2) System checks active database states. 3) System presents either a list of help options or previously placed item cards.
Alternative Flows	A1) Customer clicks outside the modal bounding box to dismiss the overlay interaction.
Exceptions	E1) Connection timeout.
Priority	Medium
Frequency of Use	High
Business Rules	Order history view only pulls records that match the current table session context..

UC2: Staff Workflow & Resolution Management

UC-2.1 Process Incoming Order Status	
Description	Staff reviews newly broadcasted pending orders, updates progress, and dispatches them to the kitchen.
Trigger	Staff opens the Merchant Dashboard -> New Orders tab.
Preconditions	Staff is authenticated and authorized; real-time network layer is connected.
Postconditions	Order tracking record state shifts; active workspace logs the processed action.
Normal Flows	1) Staff opens order management. 2) System highlights incoming orders with red indicators. 3) Staff reviews details and clicks “Accept” & “Notify Kitchen” 4) System updates status to "In Progress" and transfers the card to the preparation tab.
Alternative Flows	A1) Staff rejects the order by pressing “Cancel” if criteria are unfulfilled.
Exceptions	E1) Concurrency collision: order already approved by another worker -> refresh layout. E2) Database error.
Priority	High
Frequency of Use	Medium
Business Rules	Only authenticated merchant accounts can alter system-wide order status states.

Assumptions	Active logging tracks staff_id, event_type, target_id, and time tokens.
--------------------	---

UC-2.2 Resolve Customer Assistance Alerts

Description	Staff views and dismisses critical tableside notifications triggered by seated customers.
Trigger	Critical pop-up warning "THÔNG BÁO KHẨN" overlays the staff dashboard view.
Preconditions	An active assistance request flag is logged inside the active message table.
Postconditions	Alert modal disappears; table status state resets; staff addresses the customer's request.
Normal Flows	1) System detects a live request and prompts an overlay modal detailing Table ID and target reason (e.g., [TÍNH TIỀN]). 2) Staff acknowledges text contents. 3) Staff hits the confirmation switch "ĐANG ĐẾN XỬ LÝ " (On My Way). 4) System clears the notification flag and returns the app
Alternative Flows	None.
Exceptions	E1) Data synchronization delay -> alert fails to close -> trigger automatic list polling.
Priority	High
Frequency of Use	Medium
Business Rules	Urgent assistance alerts possess absolute layout rendering hierarchy and block alternative actions until closed.
Assumptions	Merchant terminal app is continually running in an active foreground mode.

UC3: Menu & Catalog Management

UC-3.1 Manage Menu Items	
Description	Admin/Manager performs Add/Edit/Delete (or Deactivate)/Search/Display operations for food and beverage items in the digital catalog.
Trigger	Admin opens Menu Management -> Items.
Preconditions	Admin is authenticated and authorized; system and database are available.
Postconditions	Menu item data is updated according to the selected action; the customer-facing e-Menu list is instantly refreshed; audit logs are recorded.
Normal Flows	1) Admin accesses Menu Item management screen. 2) System displays the current list of items and available management actions. 3) Admin performs a management action 4) System validates data formats and pricing constraints, then applies and saves changes.
Alternative Flows	A1) Admin uses search keyword or filters by category (e.g., Coffee, Smoothie) to narrow down items.

	A2) Admin cancels an in-progress add/edit item form.
Exceptions	E1) Duplicate item code or item name. E2) Negative or invalid price format -> system prompts correction. E3) System/DB error.
Priority	High
Frequency of Use	High
Business Rules	Item code must be unique; required fields (Name, Price, Category) must be provided; pricing cannot be less than zero.
Assumptions	Item details include at least item code, name, price, image url, availability status (In stock/Out of stock), and updated metadata.

UC-3.2 Manage Menu Categories	
Description	Admin/Manager performs Add/Edit/Delete/Assign/Search/Display operations for menu categories (e.g., Coffee, Smoothie, Fruit Juice) to structure the e-Menu.
Trigger	Admin opens Menu Management -> Categories.
Preconditions	Admin is authenticated and authorized.
Postconditions	Category database structure is updated; customer interface updates the filter tabs accordingly.
Normal Flows	1) Admin opens category management screen. 2) Admin performs a management action (creates a new category, deactivates an old one, or reorders the list). 3) Admin saves changes.
Alternative Flows	A1) Admin creates a new category based on an existing template or duplicates an operational group structure.
Exceptions	E1) Duplicate category identifier/name. E2) Deletion blocked if the category currently contains active items linked to it -> system suggests deactivation instead.
Priority	Medium
Frequency of Use	Medium
Business Rules	Category identifier must be unique; each category should map to valid active menu structures.
Assumptions	Category data includes category code, name, display order, status, and metadata.

UC-3.3 Manage Tables & QR Codes	
Description	Admin performs Add/Edit/Delete/Assign/Search/Display operations for physical table records and generates distinct QR codes mapped to each table.
Trigger	Admin opens System Management -> Tables & QR

	Codes.
Preconditions	Admin is authenticated and authorized.
Postconditions	Table data is updated; target QR codes are uniquely generated, refreshed, or mapping files are downloaded; audit logs are recorded.
Normal Flows	1) Open table management screen. 2) Perform a management action (e.g., create a new table identifier "BÀN 06"). 3) System automatically generates a unique tokenized URL and corresponding QR code graphic for the table. 4) Save changes and download the QR vector/image file.
Alternative Flows	A1) Configure sorting order, floor/area filters (e.g., Ground Floor, Rooftop), or batch-generate QR codes for multiple tables.
Exceptions	E1) Duplicate table number/identifier code. E2) Database/QR generation API timeout.
Priority	High
Frequency of Use	Medium
Business Rules	Table code must be unique within the restaurant layout; only active tables can receive digital order data via the customer interface.
Assumptions	Table data includes table_id, table_number, zone_location, current_occupancy_status, and QR_code_url metadata.

UC-3.4 Manage Combo / Menu Structure

Description	Admin performs Add/Edit/Delete/Assign/Search/Display operations for complex menu structures, organized by promotional Combos or custom recommendation groups.
Trigger	Admin opens Menu Management -> Combo & Menu Structure.
Preconditions	Admin is authenticated; valid individual food/beverage items exist in the database (from UC-3.1).

Postconditions	Custom structures, discount mappings, and recommendation metrics are updated and stored inside the active catalog.
Normal Flows	1) Select target area, time slot, or promotion version. 2) Open structure/combo editor screen. 3) Arrange categories and conditional groups (e.g., Choose 1 Coffee + 1 Pastry). 4) Assign specific menu items into the combo container. 5) Save changes.
Alternative Flows	A1) Reorder menu display hierarchies or modify group priority values.
Exceptions	E1) Constraint violations (e.g., setting a combo price higher than the sum of individual items).
Priority	High
Frequency of Use	High during seasonal menu/promotion updates.

Business Rules	Menu structures must comply with catalog structural rules; only active and in-stock menu items can be assigned into visible combos.
Assumptions	The system supports multi-item grouping and dynamic price calculation overrides for combo schemas.

UC-3.5 Manage Store Configuration & Tax Framework	
Description	Admin performs Add/Edit/Delete/Assign/Search/Display operations for global store rules, service charge constraints, and tax framework standards.
Trigger	Admin opens System Settings -> Configuration & Tax Framework.
Preconditions	Admin is authenticated and authorized.
Postconditions	Operational standards, VAT rules, and global store service fees are updated and versioned.
Normal Flows	1) Open framework management. 2) Configure standard system parameters (e.g., set VAT to 10%, apply 5% weekend service fee). 3) Save.
Alternative Flows	A1) Compare framework operational rules across different seasonal setups (if supported).
Exceptions	E1) Conflicts with active, processing table bills or pre-calculated order values.
Priority	High
Frequency of Use	Medium
Business Rules	Framework adjustments must not invalidate historical transaction summaries or completed order invoices; data versioning is recommended.
Assumptions	Framework structure includes tax percentages, extra fee rules, currency codes, and applicability time ranges.

UC-3.6 Copy Menu Catalog	
Description	Admin copies an entire existing menu schema/catalog to quickly create a new version for a different seasonal event, shift, or holiday theme.
Trigger	Admin selects "Copy Menu" on a target source catalog structure.
Preconditions	Source menu data exists; Admin possesses proper catalog duplicative permissions.
Postconditions	A new cloned menu version is generated for the target event/season container.
Normal Flows	1) Select source menu schema. 2) Choose target seasonal shift, holiday code, or active timeline. 3) Confirm transfer scope. 4) System duplicates data models.

	5) Admin reviews the cloned catalog.
--	--------------------------------------

Alternative Flows	A1) Copy the complete menu layout architecture without duplication of base pricing lists if store pricing strategies differ.
Exceptions	E1) Target timeline or event container already holds an active menu version -> system prompts user for conflict resolution.
Priority	Medium
Frequency of Use	Medium
Business Rules	Copy must preserve data integrity; edits on the new version must not affect the source.
Assumptions	Menu catalog supports versioning by shifts, seasons, or promotional events.

UC-3.7 Display/Query Menu Catalog	
Description	Admin/Staff/Customer views the menu catalog structure based on category, event session, or menu version.
Trigger	Admin/Staff/Customer opens the main e-Menu or Catalog view interface.
Preconditions	Menu catalog data exists; viewing access is allowed.
Postconditions	The selected menu layout and item details are displayed correctly on the target device.
Normal Flows	1) User selects category/session/version. 2) System loads menu structure, categories, and items. 3) Display details (names, descriptions, pricing, image elements).
Alternative Flows	A1) Export Menu data schema to PDF/Excel file (if enabled for Admin/Staff).
Exceptions	E1) Missing or invalid selected menu version -> system falls back to default active catalog.
Priority	Medium
Frequency of Use	High
Business Rules	Customer sees only approved/active versions and available items; hidden elements are restricted to administrative views.
Assumptions	Filters by category/session/availability status are available.

UC4: Order Bill & Revenue Calculation

UC-4.1 Set Up Menu Item Pricing & Tax Unit

Description	Admin configures the base price, active discount rules, and tax calculation basis (VAT, Service Charges) for billing formulas according to store regulations.
Trigger	Admin opens Finance -> Pricing & Fee Setup.
Preconditions	Admin is authenticated and authorized.
Postconditions	Base item prices, fee units, and tax calculation formulas are stored; audit logs are recorded.
Normal Flows	1) Open settings screen. 2) Input unit price updates, tax rules, or special surcharge percentages. 3) Validate mathematical ranges. 4) Save settings.
Alternative Flows	A1) Configure pricing modifiers based on specific table categories or VIP membership statuses if supported..
Exceptions	E1) Invalid input ranges (e.g., negative tax values) or business rule conflicts.
Priority	Medium

Frequency of Use	Medium
Business Rules	Tax frameworks must comply with legal standards; active modification of base pricing must instantly apply to newly created orders without changing already locked/completed
Assumptions	Core system calculations utilize standard double-entry precision formats to avoid currency rounding errors. .

UC-4.2 Display Bill & Calculation	
Description	Admin/Staff/Customer views bill calculation information specific to a Table ID, active order session, or completed transaction.
Trigger	Admin/Staff/Customer opens View Bill/Checkout for a selected Table/Order.
Preconditions	Pricing units/tax rules are configured; view access is allowed.
Postconditions	Bill calculation information is displayed accurately.
Normal Flows	1) Select table/order session. 2) System retrieves/calculates total bill amount. 3) Display breakdown (subtotal, applied VAT/surcharges, and active discounts).
Alternative Flows	A1) Export bill/receipt information (if enabled for Admin/Staff accounts).
Exceptions	E1) Missing menu pricing configuration or failed item mapping for selection.
Priority	Medium
Frequency of Use	Medium
Business Rules	Displayed bill must align with the active, approved tax and item catalog configuration.
Assumptions	Calculation uses item quantities and configured base unit prices.

ASR-01: Maintainability and Modifiability

- Type: Quality Attribute
- Description: The system must support frequent updates to menu items, localized pricing adjustments, and seamless integration of third-party payment gateways (e.g., MoMo, VNPay) with minimal impact on core transactional workflows..
- Architectural Impact: This requirement drives the adoption of a layered architecture layout, where responsibilities are clearly separated into presentation, service, domain, and data access layers. Such separation allows API integration changes and payment gateway webhooks to be localized within specific domain layers, improving maintainability and reducing coupling between the catalog and the checkout components.

ASR-02: Real-Time Order Synchronization & Notification

- Type: Functional / Quality Attribute
- Description: The system must support immediate, real-time transmission of tableside order placements and urgent assistance alerts from client browsers directly to the merchant staff terminal dashboard without page reloads.
- Architectural Impact: This requirement influences the communication protocol layer, driving the implementation of an asynchronous event-driven model (such as WebSockets or Server-Sent Events) alongside a stateless backend. Centralized event handling within the monolithic architecture guarantees instant alert propagation to active staff sessions and ensures consistent session state management across multiple physical tables.

ASR-03: Security and Data Validation Integrity

- Type: Quality Attribute
- Description: The system must strictly secure all transactional calculations, preventing unauthorized client-side parameters or malicious modifications to raw item prices during checkout operations. must maintain strong data consistency and integrity
- This requirement directly dictates the reinforcement of isolated business validation logic within the core service layer. Every inbound table submission payload triggers mandatory server-side cross-referencing against the trusted database pricing engine before a record transitions to active preparation queues, ensuring absolute system integrity.

4. Conclusion & Reflection

The requirements elicitation phase for the QR Code Food Ordering System successfully established the project scope through well-defined Functional, Non-Functional, and Architecturally Significant Requirements. The UML Use Case Diagram offers a clear visualization of system interactions across key dining actors and highlights the essential workflows involved in browsing menus, managing carts, and processing table orders.

The identified ASRs—particularly those related to Modifiability, Security, and Scalability strongly guide the need for a structured architectural approach such as the Layered Architecture. These requirements ensure that the system's design in Lab 2: Layered Architecture Design (Logical View) will support maintainable payment gateway updates, secure server-side transaction validation, and efficient performance during peak restaurant dining periods. This foundation provides a sustainable architectural direction for subsequent development phases.